GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 1 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 2	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Integers - Identification of integers Integers - Representatio n of integers on number line

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define integers and distinguish them from non-integers - Identify positive integers, negative integers and zero in different situations - Appreciate the use of integers in daily life situations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we identify integers in real life situations?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 1 - Thermometer s - Number cards - Charts with integers - Master Mathematics Grade 8, pg. 2 - Manila paper - Rulers - Markers - Number lines

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and find readings of thermometers showing positive and negative values - Classify numbers as integers or non-integers - Use real-life situations like floors above and below ground to represent integers

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Oral questions - Written exercises

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 2 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 2	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Integers - Addition of integers on number line Integers - Subtraction of integers on number line

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the rule for adding integers on a number line - Carry out addition of integers on a number line correctly - Value the importance of addition of integers in real life

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we carry out addition of integers?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 3 - Number cards - Ground markings - Chalk - Counters - Master Mathematics Grade 8, pg. 4 - Number lines - Markers - Playground space

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Use number cards and containers for selection - Draw number lines on the ground - Jump to the right to add positive numbers - Mark and record positions after jumping

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical activities - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 3 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 2	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Integers - Combined operations on number line Integers - Application of integers using IT resources

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe the order of combined operations on integers - Perform combined addition and subtraction of integers on number line - Show confidence in solving problems involving integers

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we perform combined operations of integers?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 5 - Number lines - Temperature gauges - Real- life problem cards - Master Mathematics Grade 8, pg. 6 - Digital devices - Internet access - Integer games/apps

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice mixed operations using number lines - Solve problems involving temperature changes - Work out problems involving floors in buildings

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Problem- solving tasks - Observation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 4 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 2	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Fractions - Order of operations in fractions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify the correct order of operations in fractions (BODMAS) - Carry out combined operations on fractions accurately - Appreciate the importance of following correct order

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we use fractions in real life situations?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 8 - Fraction cards - Calculators - Charts showing BODMAS

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and use correct order of operations in fractions - Work through operations in brackets first, then multiplication/ division, then addition/subtr action - Copy operations on number cards and solve

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Class activities - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 5 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 2	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Fractions - Operations on fractions from shopping activities Fractions - Word problems involving fractions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain how fractions are used in shopping and trading - Work out fraction operations from shopping activities - Show responsibility in applying fractions to real situations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Where do we apply combined operations on fractions?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 9 - Shopping lists - Price tags - Play money - Fraction pieces - Master Mathematics Grade 8, pg. 10 - Word problem cards - Fraction charts - Measuring tools

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and carry out operations on fractions from shopping and other real-life cases - Role- play shopping scenarios - Solve problems involving sharing and distribution

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Problem- solving - Practical activities - Written assignments

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 6 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 3	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Fractions - Games and IT activities on fractions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe different games involving fractions - Use IT devices for learning operations on fractions and play games - Enjoy learning about fractions

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How can we make learning fractions more interesting?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 11 - Tablets/comp uters - Internet access - Fraction games

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Play games of operations on fractions using IT devices or other resources - Engage in drag and drop activities online - Use fraction apps for practice

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Game performance - Digital assessment

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 7 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 3	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Fractions - Mixed practice on combined operations

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Recall the order of operations in fractions - Solve complex combined fraction operations proficiently - Show confidence in working with fractions

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What strategies help us solve complex fraction problems?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 12 - Exercise books - Fraction worksheets - Group work materials

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice solving mixed fraction problems - Work in groups on challenging fraction tasks - Present solutions to the class

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Group presentations - Peer assessment

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 8 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 3	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Fractions - Mixed practice on combined operations

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Recall the order of operations in fractions - Solve complex combined fraction operations proficiently - Show confidence in working with fractions

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What strategies help us solve complex fraction problems?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 12 - Exercise books - Fraction worksheets - Group work materials

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice solving mixed fraction problems - Work in groups on challenging fraction tasks - Present solutions to the class

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Group presentations - Peer assessment

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 9 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 3	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Fractions - Application and reflection

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Discuss various applications of fractions in daily life - Demonstrate mastery of fraction operations - Promote use of fractions in real life situations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How have fractions helped us in our daily lives?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 13 - Portfolio materials - Reflection journals

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss with peers, parents or guardians about areas where fractions are applied - Share real-life experiences involving fractions - Compile a portfolio of fraction work

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Portfolio assessment - Oral presentations - Self- assessment

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 10 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 3	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Decimals - Conversion of fractions to decimals

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the relationship between fractions and decimals - Convert fractions to decimals using different methods - Appreciate the connection between fractions and decimals

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we convert fractions to decimals?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 13 - Conversion charts - Calculators - Place value charts

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice converting fractions to decimals using equivalent fractions with denominators as powers of 10 - Use long division method to convert fractions - Complete conversion tables

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Oral questions - Class activities

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 11 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 4	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Decimals - Identifying and converting recurring decimals

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define recurring and non-recurring decimals - Identify recurring decimals and convert them to fractions correctly - Show interest in working with recurring decimals

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we identify and work with recurring decimals?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 15 - Decimal cards - Number cards - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and classify non- recurring and recurring decimals - Indicate recurring digits using dot notation - Practice converting recurring decimals to fractions using algebraic method

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Practical exercises - Observation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 12 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 4	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Decimals - Rounding off decimals to decimal places

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - State the rules for rounding off decimals - Round off decimal numbers to required decimal places accurately - Value accuracy in rounding decimals

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we round off decimals correctly?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 19 - Place value charts - Decimal number cards - Rounding worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and round off decimal numbers to required decimal places - Practice rounding to 1, 2, 3 decimal places - Use place value charts to understand rounding

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written assignments - Oral questions - Class tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 13 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 4	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Decimals - Expressing numbers in significant figures

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the meaning of significant figures - Write decimal and whole numbers to given significant figures - Show precision in expressing numbers

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What are significant figures and why are they important?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 21 - Number charts - Worksheets - Scientific calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Write decimal and whole numbers to given significant figures - Discuss rules for identifying significant figures - Practice expressing numbers to different significant figures

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Practical exercises - Observation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 14 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 4	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Decimals - Expressing numbers in standard form

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define standard form notation A × 10■ - Write numbers in standard form correctly and convert them back - Appreciate the use of standard form for very large and small numbers

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we express numbers in standard form?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 23 - Standard form cards - Calculators - Charts

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Write numbers in standard form on learning materials such as cards or charts - Practice expressing very large and very small numbers - Understand the power of 10 notation

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Oral questions - Class activities

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 15 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 4	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Decimals - Combined operations on decimals

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify the correct order of operations for decimals - Work out combined operations on decimals systematically - Show confidence in solving decimal problems

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we perform combined operations on decimals?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 24 - Operation cards - Calculators - Worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Work out combined operations on decimals in the correct order - Practice problems involving brackets, multiplication, division, addition and subtraction - Solve complex decimal calculations

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Problem- solving - Observation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 16 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 5	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Decimals - Application of decimals to real life

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify situations where decimals are used in daily life - Apply decimals to solve practical problems - Promote use of decimals in daily activities

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Where do we use decimals in our daily lives?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 26 - Real-life problem cards - Measuring instruments - Price lists

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and apply decimals to real life cases - Solve problems involving money, measurement s, temperature - Work with real-life scenarios

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Practical tasks - Written assignments - Oral presentations

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 17 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 5	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Decimals - Application of decimals to real life

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify situations where decimals are used in daily life - Apply decimals to solve practical problems - Promote use of decimals in daily activities

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Where do we use decimals in our daily lives?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 26 - Real-life problem cards - Measuring instruments - Price lists

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and apply decimals to real life cases - Solve problems involving money, measurement s, temperature - Work with real-life scenarios

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Practical tasks - Written assignments - Oral presentations

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 18 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 5	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Decimals - Games and digital activities

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain how digital games enhance learning of decimals - Use IT devices to play games involving decimals - Enjoy learning about decimals through interactive activities

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How can technology enhance our understanding of decimals?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 27 - Digital devices - Decimal games/apps - Internet access

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Play games of operations on decimals using IT or other materials - Use decimal apps and online games - Engage in interactive decimal activities

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Game performance - Participation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 19 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 5	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Squares and Square Roots - Reading squares from tables

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain how to read mathematical tables for squares - Work out squares of numbers between 1.0 and 9.999 from tables - Show accuracy in using mathematical tables

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What are squares of numbers?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 29 - Mathematical tables - Number cards - Worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Read and write the squares of numbers from tables - Practice locating numbers in the table and reading their squares - Work through examples using Table 1.3

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Practical exercises - Written tests - Observation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 20 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 5	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Squares and Square Roots - Squares of large numbers

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe the method for finding squares of numbers above 10 - Work out squares of numbers above 10 using standard form and tables - Demonstrate systematic approach in calculations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we find squares of numbers greater than 10?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 33 - Mathematical tables - Standard form charts - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice finding squares of numbers above 10 using standard form method - Convert numbers to standard form A × 10■ - Calculate squares and express in ordinary form

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Class activities - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 21 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 6	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Squares and Square Roots - Squares of numbers less than 1

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the process for squaring decimal numbers less than 1 - Find squares of decimal numbers less than 1 using tables - Show precision in working with small numbers

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we find squares of numbers less than 1?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 35 - Mathematical tables - Decimal cards - Worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice finding squares of numbers less than 1 - Use standard form with negative powers of 10 - Apply systematic method for calculations

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Practical exercises - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 22 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 6	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Squares and Square Roots - Reading square roots from tables

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain how to read square root tables - Work out square roots of numbers from 1 to 99.99 using tables - Appreciate the relationship between squares and square roots

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Where do we apply square roots in real life?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 37 - Mathematical tables - Square root charts - Number cards

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Read and write the square roots of numbers from tables - Practice using Table 1.4 for square roots - Add values from the ADD column correctly

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written assignments - Oral questions - Class tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 23 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 6	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Squares and Square Roots - Reading square roots from tables

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain how to read square root tables - Work out square roots of numbers from 1 to 99.99 using tables - Appreciate the relationship between squares and square roots

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Where do we apply square roots in real life?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 37 - Mathematical tables - Square root charts - Number cards

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Read and write the square roots of numbers from tables - Practice using Table 1.4 for square roots - Add values from the ADD column correctly

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written assignments - Oral questions - Class tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 24 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 6	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Squares and Square Roots - Square roots of large numbers

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe the method for finding square roots of numbers 100 and above - Find square roots of numbers 100 and above using tables - Show systematic approach in calculations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we find square roots of numbers above 100?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 39 - Mathematical tables (Tables 1.4 & 1.5) - Worksheets - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice finding square roots of numbers above 100 - Use standard form method - Work with both Table 1.4 and Table 1.5 appropriately

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Practical work - Observation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 25 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 6	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Squares and Square Roots - Using calculators for squares and square roots

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify the square and square root functions on a calculator - Work out squares and square roots using a calculator correctly - Appreciate the efficiency of using calculators

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do calculators help us find squares and square roots?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 42 - Scientific calculators - Digital devices - Comparison worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice working out squares and square roots using a calculator - Compare calculator results with table results - Use IT devices or other materials to play square and square root games

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Practical exercises - Observation - Written tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 26 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 7	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Identifying rates

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define rate as a quotient relationship between two quantities - Identify rates in different real-life situations - Appreciate the use of rates in daily life

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we use rates in real life situations?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 44 - Stopwatches - Rate cards - Mobile phones (for demonstration)

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Time while doing different activities such as calling using different mobile service providers - Role play activities and note time taken - Record and compare rates

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Oral questions - Practical activities

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 27 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 7	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Working out rates

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the method for calculating rates - Calculate rates from given information accurately - Show precision in rate calculations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we calculate rates from given information?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 46 - Timers - Measuring tools - Rate worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Carry out activities to determine rates - Calculate rates per unit time or quantity - Solve rate problems from real-life contexts

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Problem- solving - Class activities

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 28 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 7	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Expressing fractions as ratios

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain how to convert fractions to ratios - Express fractions as ratios in simplest form - Value precision in ratio work

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we express fractions as ratios?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 48 - Cut-out materials - Ratio cards - Counters

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Use cut outs from whole objects to relate fractions to ratios - Practice writing fractions as numerator : denominator - Simplify ratios to lowest terms

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Practical work - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 29 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 7	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Comparing ratios

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe methods for comparing two or more ratios - Compare ratios using percentage method and LCM method - Show systematic approach in comparing ratios

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we compare two or more ratios?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 50 - Comparison charts - Ratio cards - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and compare ratios from cut outs - Use LCM method to compare ratios - Express ratios as percentages for easy comparison

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Class activities - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 30 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 7	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Comparing ratios

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe methods for comparing two or more ratios - Compare ratios using percentage method and LCM method - Show systematic approach in comparing ratios

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we compare two or more ratios?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 50 - Comparison charts - Ratio cards - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and compare ratios from cut outs - Use LCM method to compare ratios - Express ratios as percentages for easy comparison

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Class activities - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 31 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 8	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Division of quantities in ratios

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the process of dividing quantities in given ratios - Divide quantities in given ratios systematically - Show fairness in sharing quantities

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we divide quantities using ratios?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 51 - Counters - Bottle tops - Sharing materials

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and share quantities of concrete objects in different ratios - Use counters or bottle tops to practice sharing - Solve sharing problems

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Practical exercises - Written assignments - Observation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 32 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 8	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Working out ratios

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify the method for finding ratios from given quantities - Work out ratios in different situations - Appreciate applications of ratios in daily life

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we determine ratios from given information?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 53 - Data cards - Real- life examples - Worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Calculate ratios from given quantities - Find missing values in ratio problems - Apply ratios to real situations

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Problem- solving - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 33 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 8	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Increase and decrease using ratios

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain how ratios show increase or decrease in quantities - Work out increase and decrease of quantities using ratios - Apply ratio changes to real situations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do ratios represent increase or decrease?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 55 - Change scenario cards - Calculators - Worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and determine increase and decrease using ratios - Use the format new : old to express changes - Solve problems involving ratio changes

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Class activities - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 34 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 8	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Percentage increase

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define percentage increase - Calculate percentage increase accurately using the formula - Show precision in percentage calculations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we calculate percentage increase?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 57 - Percentage charts - Calculators - Problem cards

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and determine percentage increase of different quantities - Use the formula: percentage change = (change/origin al) × 100% - Solve real-life percentage problems

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Practical exercises - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 35 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 8	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Percentage decrease

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define percentage decrease - Calculate percentage decrease correctly - Apply percentage decrease to real situations responsibly

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we calculate percentage decrease?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 58 - Discount cards - Price lists - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Work through percentage decrease problems - Calculate new values after percentage decrease - Solve problems involving discounts and reductions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written assignments - Problem- solving - Class tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 36 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 9	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Identifying direct proportions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define direct proportion - Identify direct proportions in real life situations - Appreciate proportional relationships in daily activities

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What is direct proportion?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 59 - Proportion charts - Real- life examples - Digital devices

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Use IT devices or other materials to explore proportions - Role play shopping activities to show direct relationships - Identify situations where increase in one leads to increase in other

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Oral questions - Practical activities

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 37 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 9	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Working out direct proportions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the unitary method for solving direct proportion - Work out direct proportions systematically - Show accuracy in direct proportion calculations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we solve direct proportion problems?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 60 - Proportion tables - Worksheets - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Complete tables showing direct proportional relationships - Calculate missing values in direct proportion - Apply direct proportion to solve problems

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Problem- solving - Class activities

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 38 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 9	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Working out direct proportions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the unitary method for solving direct proportion - Work out direct proportions systematically - Show accuracy in direct proportion calculations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we solve direct proportion problems?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 60 - Proportion tables - Worksheets - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Complete tables showing direct proportional relationships - Calculate missing values in direct proportion - Apply direct proportion to solve problems

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Problem- solving - Class activities

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 39 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 9	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Identifying indirect proportions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define indirect proportion - Identify indirect proportions in different situations - Appreciate the difference between direct and indirect proportion

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What is indirect proportion?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 62 - Hourglass - Containers - Bottle tops

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Use hourglass to show and determine indirect relationships - Identify situations where increase in one leads to decrease in other - Practice with filling containers

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical work - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 40 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 9	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Working out indirect proportions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the method for solving indirect proportion - Work out indirect proportions systematically - Show understanding of inverse relationships

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we solve indirect proportion problems?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 63 - Proportion worksheets - Calculators - Problem cards

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Complete tables showing indirect proportional relationships - Calculate values where ratios are inverted - Solve time- speed- distance problems

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Problem- solving - Written tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 41 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 10	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Numbers
Sub-strand / lesson focus: Rates, Ratio, Proportions and Percentages - Application and reflection

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Discuss various applications of ratios and proportions - Apply ratios and proportions in various real- life contexts - Promote use of ratios and proportions in real life

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do ratios and proportions help us in daily life?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 64 - Video resources - Digital devices - Portfolio materials

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Watch videos on ratios and proportions as used in daily activities - Discuss applications with parents or guardians - Reflect on learning and compile portfolio

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Portfolio assessment - Presentations - Self- assessment

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 42 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 10	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Algebraic Expressions - Factorisation of algebraic expressions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define factorisation as the reverse of expansion - Identify the highest common factor (HCF) in algebraic expressions - Appreciate the use of factorisation in simplifying expressions

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we factorise algebraic expressions?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 65 - Number cards - Algebraic expression cards - Charts

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Make three sets of cards showing algebraic expressions and their factored forms - Match cards from different rows to form equations - Discuss and identify common factors in terms - Write HCF in front of brackets and remaining factors inside

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Card matching activity - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 43 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 10	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Algebraic Expressions - Identifying like and unlike terms in factorisation

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the concept of like and unlike terms - Find common factors for different sets of terms - Show systematic approach in identifying factors

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What makes terms like or unlike in algebra?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 67 - Factor cards - Worksheets - Group work materials

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss and identify like and unlike terms - Find common factors from given sets of algebraic terms - Practice factorising expressions with numerical and variable common factors - Work in groups to factorise various expressions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Group presentations - Class activities

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 44 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 10	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Algebraic Expressions - Simplification of algebraic fractions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the process of simplifying algebraic fractions - Simplify algebraic fractions by finding LCM of denominators - Value accuracy in simplifying fractions

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we simplify algebraic expressions?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 68 - Fraction charts - LCM charts - Worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss like and unlike terms in algebraic fractions - Find LCM of denominators in algebraic fractions - Combine fractions with different denominators - Practice simplifying complex algebraic fractions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Practical exercises - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 45 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 10	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Algebraic Expressions - Simplification of algebraic fractions

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the process of simplifying algebraic fractions - Simplify algebraic fractions by finding LCM of denominators - Value accuracy in simplifying fractions

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we simplify algebraic expressions?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 68 - Fraction charts - LCM charts - Worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Discuss like and unlike terms in algebraic fractions - Find LCM of denominators in algebraic fractions - Combine fractions with different denominators - Practice simplifying complex algebraic fractions

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Practical exercises - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 46 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 11	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Algebraic Expressions - Advanced simplification practice

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe steps for simplifying complex algebraic fractions - Simplify algebraic fractions involving multiple operations - Show confidence in working with algebraic fractions

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What strategies help us simplify complex algebraic fractions?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 69 - Practice worksheets - Real-life problem cards - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice writing fractions as single fractions - Simplify fractions with algebraic denominators - Solve problems involving algebraic fractions - Work through real-life applications

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written assignments - Class tests - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 47 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 11	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Algebraic Expressions - Using IT devices and application

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify IT resources for learning algebra - Use IT devices to work out algebra exercises and drag-drop activities - Enjoy using algebraic expressions in real life situations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How can technology enhance our understanding of algebra?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 71 - Digital devices - Internet access - Algebra apps/software

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Use IT devices to work out exercises and activities in algebra - Engage in drag and drop activities of grouping similar terms - Play online games simplifying algebraic expressions - Discuss applications with peers and parents

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Digital assessment - Participation

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 48 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 11	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Linear Equations - Forming linear equations in two unknowns

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Define linear equations in two unknowns - Form linear equations from real-life situations using two variables - Show interest in forming equations from word problems

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we solve linear equations in two unknowns?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 72 - Beam balance - Masses (500g) - Marbles - Shopping scenario cards

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Put masses on beam balance and add marbles to balance - Give letters to represent unknowns - Role play shopping activities to form equations - Write equations from balancing scenarios

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Observation - Practical activities - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 49 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 11	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Linear Equations - More practice on forming equations

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Interpret word problems involving two unknowns - Form linear equations from various real-life scenarios - Appreciate the relevance of equations in daily life

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

Where do we use linear equations in two unknowns in real life situations?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 73 - Word problem cards - Real-life scenario cards - Worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Write equations to represent ages, costs, and quantities - Form equations from perimeter problems - Create equations from problems involving animals and farming - Practice with two-digit number problems

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Problem- solving - Class activities

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 50 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 11	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Linear Equations - Solving by substitution method

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the substitution method for solving simultaneous equations - Solve linear equations in two unknowns using substitution systematically - Show precision in solving equations

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we use substitution method to solve linear equations?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 74 - Fruit pictures - Equation cards - Step-by-step charts

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Write equations from fruit vendor scenario - Name equations as (i) and (ii) - Write one variable in terms of another - Replace and simplify to find values of unknowns

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Practical exercises - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 51 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 12	Lesson: 1
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Linear Equations - Advanced practice on substitution method

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe the complete process of substitution method - Solve complex simultaneous equations by substitution - Demonstrate mastery of substitution technique

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What are the key steps in substitution method?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 75 - Practice worksheets - Real-life problem cards - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice solving equations with fractions using substitution - Work through problems involving costs and quantities - Solve problems about carpentry and furniture making - Apply substitution to number problems

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written assignments - Problem- solving - Class tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 52 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 12	Lesson: 2
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Linear Equations - Advanced practice on substitution method

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Describe the complete process of substitution method - Solve complex simultaneous equations by substitution - Demonstrate mastery of substitution technique

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

What are the key steps in substitution method?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 75 - Practice worksheets - Real-life problem cards - Calculators

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice solving equations with fractions using substitution - Work through problems involving costs and quantities - Solve problems about carpentry and furniture making - Apply substitution to number problems

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written assignments - Problem- solving - Class tests

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 53 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 12	Lesson: 3
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Linear Equations - Solving by elimination method

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Explain the elimination method for solving simultaneous equations - Solve linear equations using elimination method systematically - Appreciate the efficiency of elimination method

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do we solve equations using elimination method?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 76 - Shopping scenario cards - Elimination charts - Step- by-step guides

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Form equations from shopping scenarios (plates and cups) - Multiply equations to make coefficients equal - Subtract corresponding parts to eliminate one variable - Solve for remaining variable and substitute back

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written exercises - Practical work - Oral questions

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 54 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 12	Lesson: 4
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Linear Equations - More practice on elimination method

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Identify when to use elimination method - Solve various simultaneous equations by elimination efficiently - Show confidence in choosing appropriate methods

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

When is elimination method more suitable than substitution?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 78 - Comparison charts - Practice worksheets - Method selection guides

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Practice solving equations involving bread and tea leaves - Work through problems with different coefficients - Solve problems about costs of items - Compare elimination and substitution methods

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Written tests - Class activities - Problem- solving

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.

GRADE 8 MATHEMATICS — TERM 1

LESSON PLAN 55 | CBC | 2027

_____	Teacher: _____	TSC No.: _____
	Week: 12	Lesson: 5
_____	Time: _____	Roll: _____

STRAND AND SUB-STRAND

Strand: Algebra
Sub-strand / lesson focus: Linear Equations - Application in real-life situations

LESSON LEARNING OUTCOMES

At the end of the lesson, the learner should be able to:

Curriculum outcomes: By the end of the lesson, the learner should be able to: - Discuss various applications of linear equations in daily life - Apply linear equations to solve real-life problems involving rectangles, costs, and quantities - Recognize use of linear equations in real life

Knowledge: Explain the mathematical concepts, terms, rules and relationships stated in the curriculum outcomes above.

Skills: Demonstrate the procedures and apply the mathematical processes stated in the curriculum outcomes to relevant tasks.

Attitudes: Demonstrate accuracy, perseverance, confidence and responsible participation while engaging with the mathematical tasks.

KEY INQUIRY QUESTION

How do linear equations help us solve real-life problems?

LEARNING RESOURCES

- Master Mathematics Grade 8, pg. 79 - Video resources - Real-life scenario cards - Digital devices - Application worksheets

ORGANISATION OF LEARNING

Learners work individually, in pairs and in small groups. The teacher facilitates inclusive participation, provides appropriate support and extension, and monitors progress.

INTRODUCTION

The teacher reviews prerequisite knowledge through oral questions or a short task linked to the previous lesson. Learners share prior experiences related to the lesson focus. The key inquiry question is introduced to stimulate curiosity.

LESSON DEVELOPMENT

1. Exploration: In groups, learners are guided to: - Find sum and difference of two numbers using equations - Solve problems about rectangular flower beds - Work out problems involving hiring labourers - Apply equations to school fees and shopping scenarios - Watch videos on linear equations applications

2. Teacher facilitation: The teacher guides learners through the key mathematical ideas and models relevant examples step by step, drawing directly on the stated sub-strand and curriculum outcomes.

3. Guided practice: Learners attempt representative questions individually and in pairs. They explain their methods, compare approaches and use appropriate mathematical language, diagrams or instruments.

4. Independent application: Learners complete a short task based on the lesson focus. The teacher observes performance, identifies misconceptions and provides immediate feedback or differentiated support.

ASSESSMENT

- Portfolio assessment - Presentations - Written assignments - Self- assessment

CONCLUSION

Learners summarise the main concept or procedure, respond to the key inquiry question and identify one area they can now explain or perform. The teacher gives a brief follow-up practice task aligned to the lesson outcomes.

REFLECTION / SELF-REMARKS

Achievement: _____
Challenges observed: _____
Remedial / enrichment action: _____

Curriculum source: Approved Grade 8 Mathematics scheme of work. The strand, sub-strand, outcomes, learning experiences, inquiry question, resources and assessment wording above are drawn from the corresponding scheme entry.